It is notable that features can be built by nested patterns. e.g. A window can be rectangular in shape with two rectangular panes. If each pane contains a circle drawn over it for decorative purposes, then the windows can be identified as a rectangle made of two rectangles with a circle inside each. In such cases the successful recognition of the window would depend on the algorithm and the nodes available in the feature search tree.

Yet another hurdle in this case is the size and relative position of other objects. Suppose that the feature interpretation based search has confirmed that the object in front of the camera is a house but it also identifies a shoe in the image which happens to be larger than the house itself, then such a match would suggest that the house is actually a toy kept besides a real shoe. For the recognition algorithm to correctly peg this, it would require more than feature interpretation.

WHAT IS A FEATURE?

A feature, in terms of object recognition is a shape identifiable in a picture. Basic shapes (such as a rectangle, square, pentagon, circle etc.) too qualify as features. As you combine more than one feature together, you get more complex ones. In our case, a window is a feature and it is built by rectangles placed inside other rectangles. The identification of a feature depends on pattern recognition.can be used for image registration.

Hypothesization and testing

In this technique, various features are detected in the image first. The list of features are then searched against the list of features in known object models. The overall location of the features is also sent as a parameter for search. From the resulting set, the object model is then projected in software to decide whether the projection of the object returned as a search result would look similar to what is being seen by the camera. If the hypothetical image (projection of the object model received as a search result) is sufficiently similar to the image as seen by the camera, then the hypothesis is accepted and the object is recognised as the one received in search results.

Scale Invariant Feature Transformation (SIFT)

SIFT works by first storing key points (features) of an object and storing them in a database. This is done before SIFT is employed for object recognitions.